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Started on Wednesday, 16 October 2024, 12:59 PM

State Finished

Completed on Wednesday, 16 October 2024, 1:28 PM

Time taken 28 mins 24 secs

Marks 1.81/3.00

Grade **1.81** out of 3.00 (**60.2%**)

Information

Information

This page contains all the problems for this test. The very last problem asks you to contact the person in charge of the exam and tell him or her the 4-digit key given in the problem text. In return you will be given a 5-digit signing code which you must give as the answer to the problem.

This problem does not count towards the final score, but **tests missing this code will not count towards the final grade.**

The following rules apply:

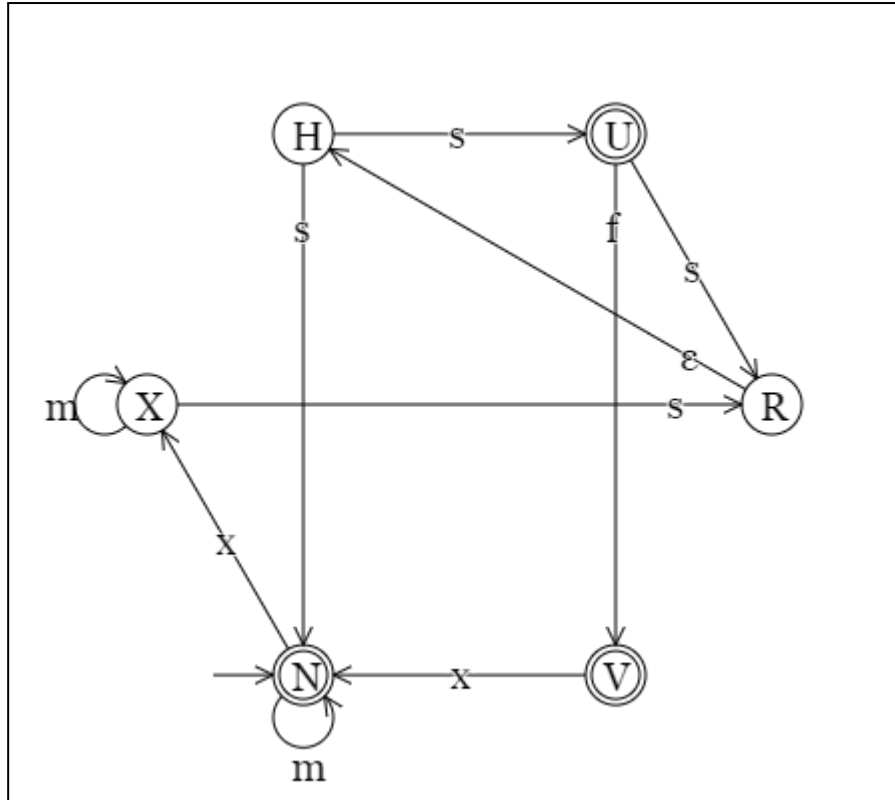
- Total time allowed: 30 minutes. The test will automatically close if time runs out.
- UiA's usual rules in regards to cheating on exams apply.

Question 1

Partially correct

Mark 0.56 out of 1.00

Translate the following DFA to a right-regular grammar.



Write the answer as a 4-tuple, separating the rule per non-terminal. Use the names of the states for their respective nonterminals.

Your answers must be syntactically correct for marking.

Terminals = {"x","m","s","f"}

Your last answer was interpreted as follows:

{x, m, s, f}

Nonterminals =

Your last answer was interpreted as follows:

$\{N, V, X, R, H, U\}$

Start symbol =

Your last answer was interpreted as follows:

N

Rules for R =

Your last answer was interpreted as follows:

$\{[R, [e, H]]\}$

Rules for V =

Your last answer was interpreted as follows:

$\{[V, [x, N]]\}$

Rules for N =

Your last answer was interpreted as follows:

$\{[N, [m, N]]\}$

Rules for X =

Your last answer was interpreted as follows:

$\{[X, [m, X]], [X, [s, R]]\}$

Rules for H =

Your last answer was
interpreted as follows:

$$\{[H, [s, N]], [H, [s, U]]\}$$

Rules for U =

Your last answer was
interpreted as follows:

$$\{[U, [f, V]], [U, [s, R]]\}$$

Question **2**

Partially correct

Mark 0.25 out of 1.00

Apply the refinement rule "Introduce Constant" (see below) to the following specification statement in order to introduce a constant T of type Nat with the value 8.

Specification statement: $\{ 18 < c \} c, x, h := c', x', h' \mid c' \neq c$

Refinement rule Introduce Constant (InC)

$\{ \text{pre} \} v, x := v', x' \mid \text{rel}$

$[=$

$\text{CON } C:T \mid C=x . \{ C=x \text{ and } \text{pre} \} v, x := v', x' \mid \text{rel}$

Fill in the correctly refined specification statement. You can copy parts of the original statement.

CON T: NAT.T:=8; $\{ 18 < c \} c, x, h, T := c', x', h', T \mid c' \neq c$

Your last answer was interpreted as follows:

CON T: NAT.T:=8; $\{ 18 < c \} c, x, h, T := c', x', h', T \mid c' \neq c$

Question **3**

Correct

Mark 1.00 out of
1.00

Let R be a relation over $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$.

We know that R is an equivalence relation and a partial order.

Moreover, $(6, 8)$ is not in R .

Select all pairs that are in R .

☐ $[1, 4]$ ☐ $[2, 5]$ ☐ $[3, 9]$ ☐ $[4, 6]$ ☐ $[5, 9]$ ☒ $[6, 6]$ ☐ $[6, 8]$ ☐ $[7, 6]$ ☐ $[7, 9]$ ☐ $[8, 3]$ ☐ $[8, 6]$ ☒ $[8, 8]$ ☐ $[9, 3]$ ☐ None of the above

Question **4**

Correct

Mark 0.00 out of
0.00

Signing code

Before closing the test you must answer this problem with a signing code given to you by the person in charge of the test.

Tests missing this signing code will be ignored and will not count towards the final score.

Key: 358

Signing code:

Your last answer was interpreted as follows:

53881

◀ [Test 1 \(topics 1-3: Introduction, Concepts, Induction, Recursion, Grammars\)](#)